

HEART FAILURE — Clinical Revision Table (ESC 2021/2023)

Bold = high-yield. HF = clinical syndrome from structural/functional cardiac abnormality → ↑intracardiac pressure and/or ↓cardiac output.

1 — CLASSIFICATION BY EJECTION FRACTION

Type	Definition (LVEF)	Criteria	Notes
HF_rEF	LVEF ≤40% (reduced) — systolic failure	Symptoms ± signs	~50% of HF. Only group where prognosis improved by treatment
HF_{mr}EF	LVEF 41–49% (mildly reduced)	Symptoms ± signs	Intermediate
HF_pEF	LVEF ≥50% (preserved) — diastolic	Symptoms ± signs + objective structural/functional abnormality (LV diastolic dysfunction, ↑filling pressure, ↑natriuretic peptides)	~50% of HF. Common in elderly/HTN/AF. Survival = HF _r EF

2 — PATHOPHYSIOLOGY & CAUSES

Pathophysiology	Myocardial damage → ↓LV function → activation of neurohormonal systems (SNS, RAAS, ↑AVP/ADH, ↑natriuretic peptides) → vasoconstriction (↑afterload), Na/water retention (↑preload), hypertrophy/remodelling → vicious cycle → congestion + ↓output. CO = HR × SV; SV depends on preload, afterload, contractility
Causes	Coronary artery disease (commonest) , hypertension, valvular disease, cardiomyopathies (dilated/hypertrophic/restrictive), arrhythmias (AF), congenital, myocarditis, pericardial disease. High-output: anaemia, thyrotoxicosis, AV fistula, Paget's. Toxins: alcohol, chemo (anthracyclines), drugs

3 — SYMPTOMS & SIGNS (ESC Table 6)

	High-yield	Other
Typical symptoms	Breathlessness, orthopnoea, paroxysmal nocturnal dyspnoea (PND), reduced exercise tolerance, fatigue, ankle swelling	Less typical: nocturnal cough, wheeze, bloating, anorexia, confusion (elderly), palpitation, dizziness, syncope, bendopnoea
More specific signs	↑JVP, hepatojugular reflux, third heart sound (gallop), laterally displaced apex beat	Less specific: weight gain >2kg/wk, cachexia, murmur, peripheral oedema, pulmonary crackles, pleural effusion, tachycardia, hepatomegaly, ascites, cold extremities, oliguria, narrow pulse pressure

4 — SEVERITY (NYHA) & STAGES

NYHA	Functional class	Stage	ACC/AHA stage
NYHA I	No limitation — ordinary activity no symptoms	Stage A	At risk (HTN, DM, CAD), no structural disease/symptoms
NYHA II	Slight limitation — symptoms on ordinary activity	Stage B	Structural disease, NO symptoms (pre-HF)
NYHA III	Marked limitation — symptoms on less than ordinary activity	Stage C	Structural disease + current/past symptoms
NYHA IV	Symptoms at rest / any activity → discomfort	Stage D	Advanced/refractory HF, needs specialised intervention

5 — EPIDEMIOLOGY & PROGNOSIS

Epidemiology	Incidence Europe ~3/1000 person-yrs. Prevalence ~1–2% adults, >10% if ≥70 yrs. ~50% HF _r EF, ~50% HF _p EF/HF _{mr} EF
Prognosis	1-yr mortality 20%, 5-yr mortality 53% (all types). Poor QOL. Improvement confined to HF _r EF. Women better survival than men

6 — DIAGNOSIS

Chronic HF	All patients (Class I): BNP/NT-proBNP, 12-lead ECG, transthoracic echo, chest X-ray. Routine bloods (FBC, U&E, glucose/HbA1c, TFT, lipids, iron studies)
Acute HF	ECG, pulse oximetry, echo, labs, CXR, lung ultrasound + natriuretic peptides . Rule OUT: BNP <100 / NT-proBNP <300 / MR-proANP <120 pg/mL. Confirm: BNP ≥100 / NT-proBNP ≥300 / MR-proANP ≥120 → comprehensive echo

7 — ACUTE HEART FAILURE (in-hosp mortality 4–10%)

4 presentations	1. Acute decompensated HF (commonest) 2. Acute pulmonary oedema (severe dyspnoea, orthopnoea, crackles) 3. Isolated RV failure (↑JVP, no congestion) 4. Cardiogenic shock (hypoperfusion, ↓BP, oliguria, cold)
Find precipitant (CHAMPIT)	C =acute Coronary syndrome · H =Hypertension emergency · A =Arrhythmia · M =Mechanical cause · P =Pulmonary embolism · I =Infection · T =Tamponade
Management	Oxygen/ventilatory support (if hypoxic), IV loop diuretics (furosemide) for congestion, vasodilators if ↑BP, inotropes/vasopressors if shock (dobutamine, dopamine, milrinone, levosimendan, norepinephrine, epinephrine), treat the CHAMPIT cause, MCS if refractory

8 — CHRONIC HF MANAGEMENT

	Therapy	Note
HFrEF — 4 pillars	1. ACEi/ARB or ARNI (sacubitril-valsartan) · 2. Beta-blocker · 3. MRA (spironolactone/eplerenone) · 4. SGLT2 inhibitor (dapagliflozin/empagliflozin) — all Class I. + Diuretic for fluid retention (symptom relief). Devices: ICD, CRT	Foundational therapy ↓mortality
HFmrEF	Diuretics for congestion; SGLT2 inhibitor (dapa/empagliflozin); treat CV + non-CV comorbidities	
HFpEF	Diuretics for fluid retention; SGLT2 inhibitor ; treat underlying CV (HTN, AF) + non-CV comorbidities	

9 — KEY DRUG DOSES (HFrEF)

Class	Dose (start → target)	Monitor
ACEi	Ramipril 2.5mg→10mg/d · Enalapril 2.5mg BD→10–20mg BD · Lisinopril 2.5–5→20–35mg/d	Monitor BP, U&E, renal function
ARNI	Sacubitril/valsartan 24/26mg BD → 97/103mg BD	Replaces ACEi (36h washout)
Beta-blocker	Bisoprolol 1.25→10mg/d · Carvedilol 3.125→25mg BD	Start low, go slow
MRA	Spironolactone / eplerenone 25mg → 50mg/d	Monitor K+, renal (avoid if K>5/eGFR<30)
SGLT2 inhibitor	Dapagliflozin / empagliflozin 10mg/d	Euglycaemic DKA risk; genital infection

10 — ADVANCED HF & MECHANICAL CIRCULATORY SUPPORT (MCS)

Advanced HF	Persistent severe symptoms despite maximal therapy → consider MCS / transplant
MCS indications	BTD Bridge to decision (ECMO/Impella in shock) · BTC Bridge to candidacy · BTT Bridge to transplantation (LVAD/TAH) · BTR Bridge to recovery · DT Destination therapy (long-term LVAD if transplant-ineligible)